Diferença de Gaussianas

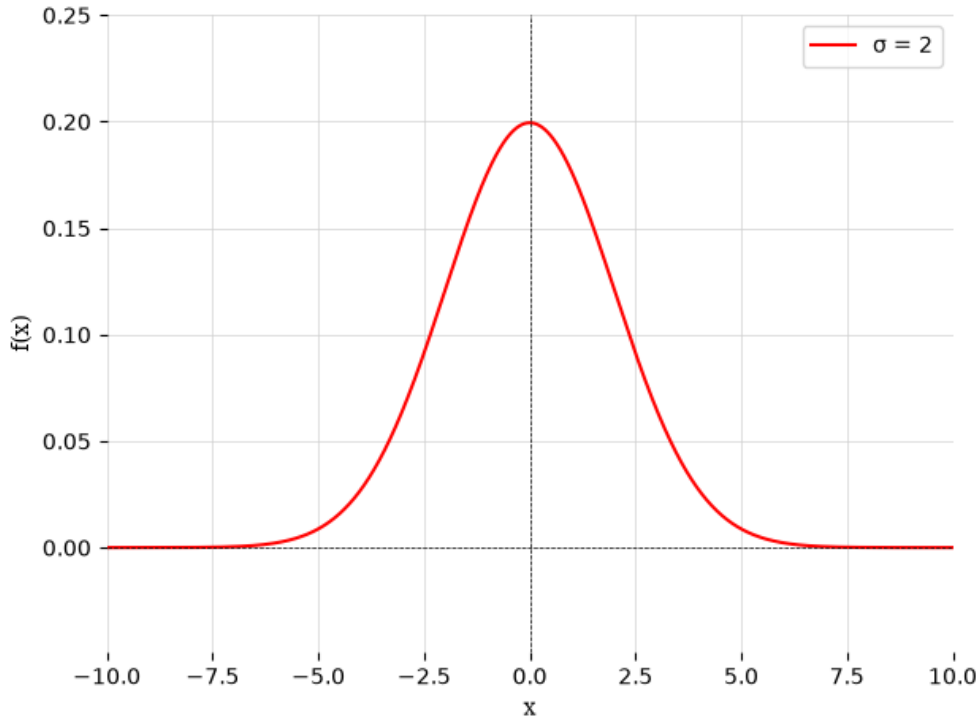
Tiago Maia de Lacerda Brasil

PPGC - IC/UFF

2026

Gaussian blur

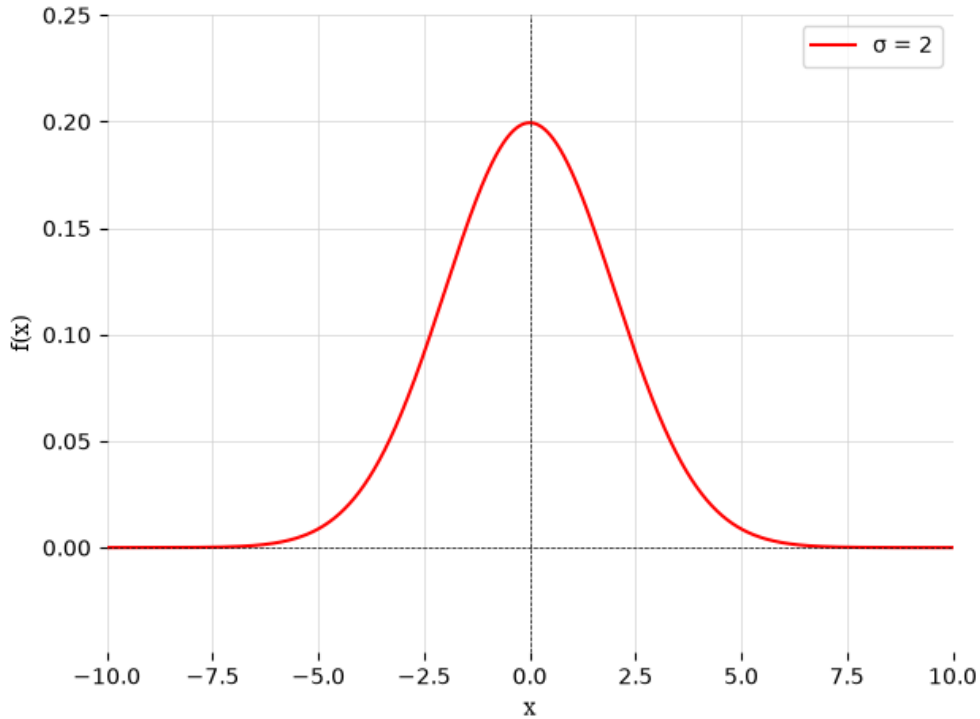
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Gaussian blur



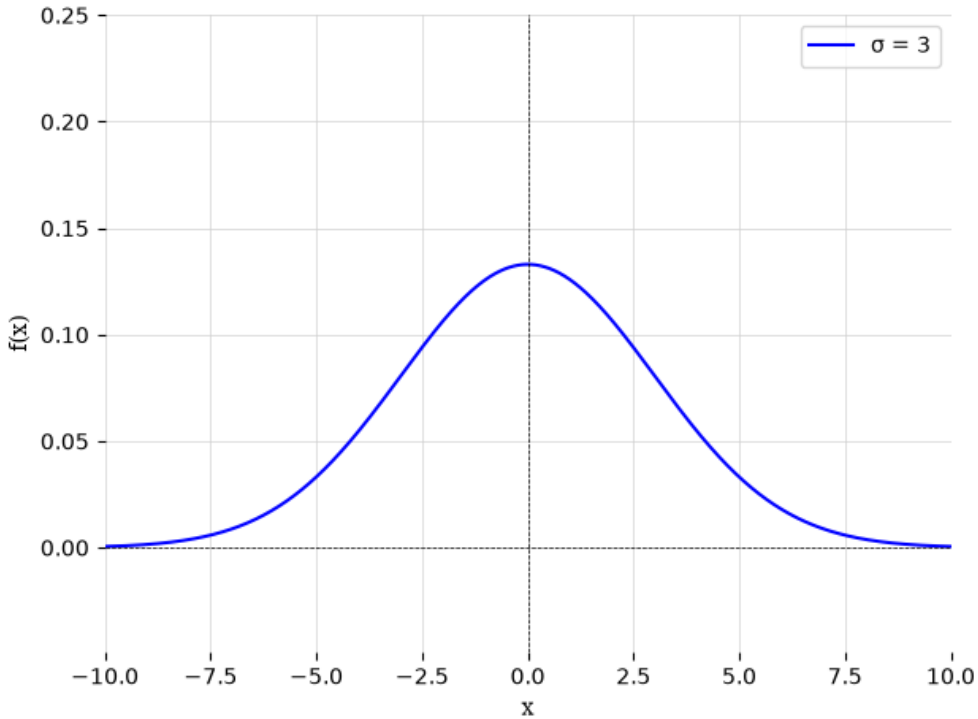
$\sigma = 2$



Gaussian blur



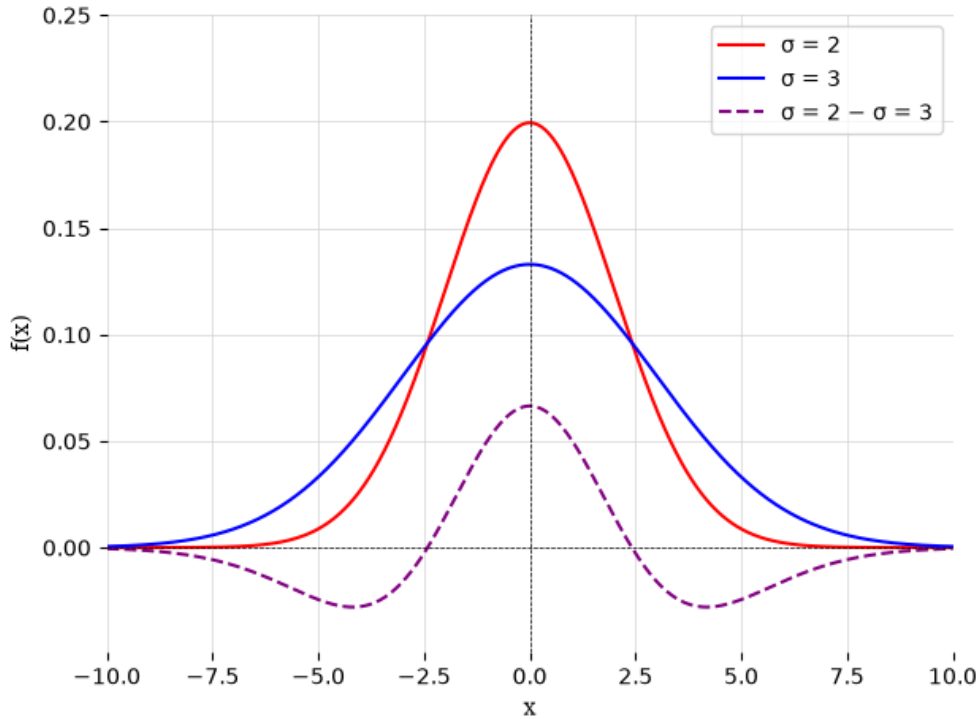
$\sigma = 3$



Gaussian blur



$(\sigma = 2) - (\sigma = 3)$



Sobel operator



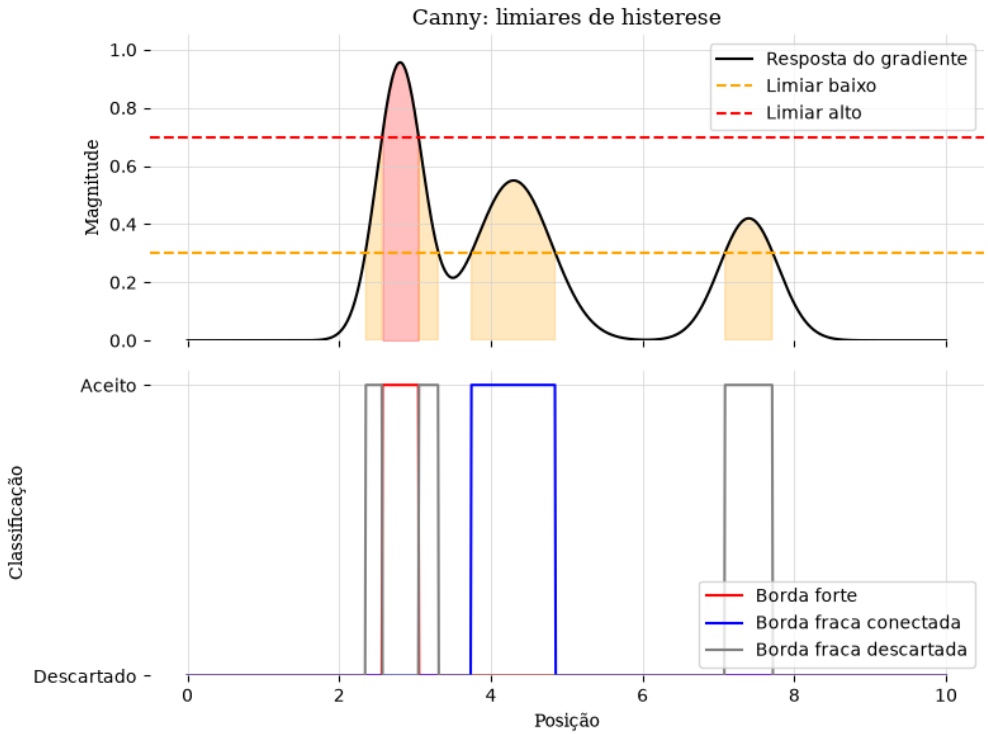
$$I = \sqrt{G_x^2 + G_y^2}$$

$$G_x = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

$$G_y = \begin{array}{|c|c|c|} \hline -1 & -2 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

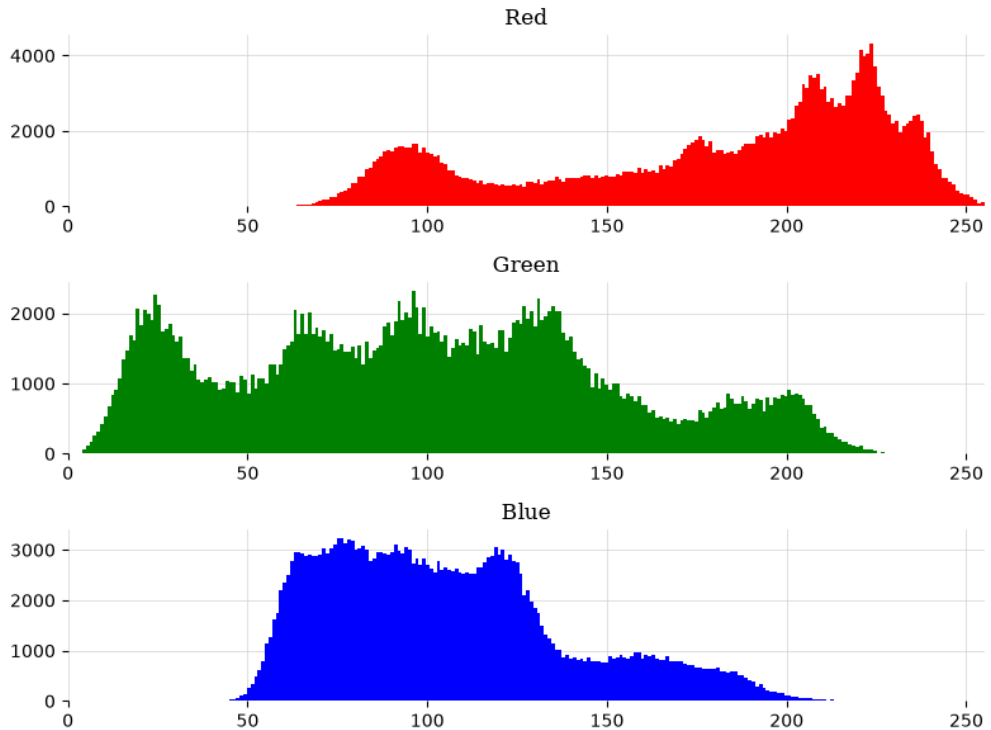
Canny edge detector





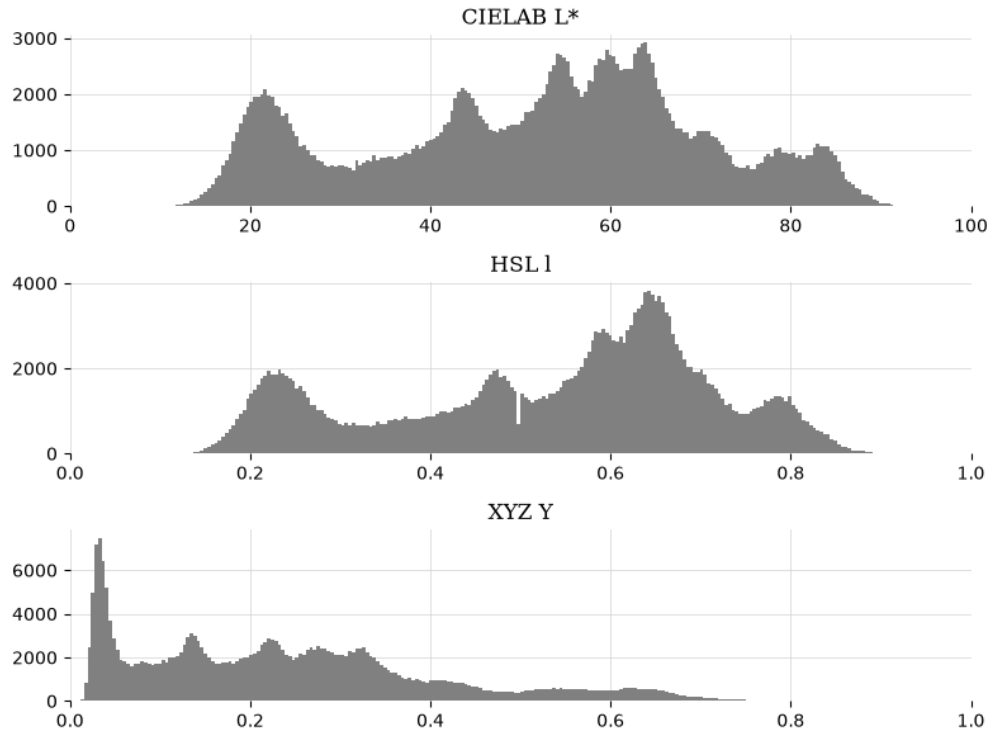
Espaços de cores



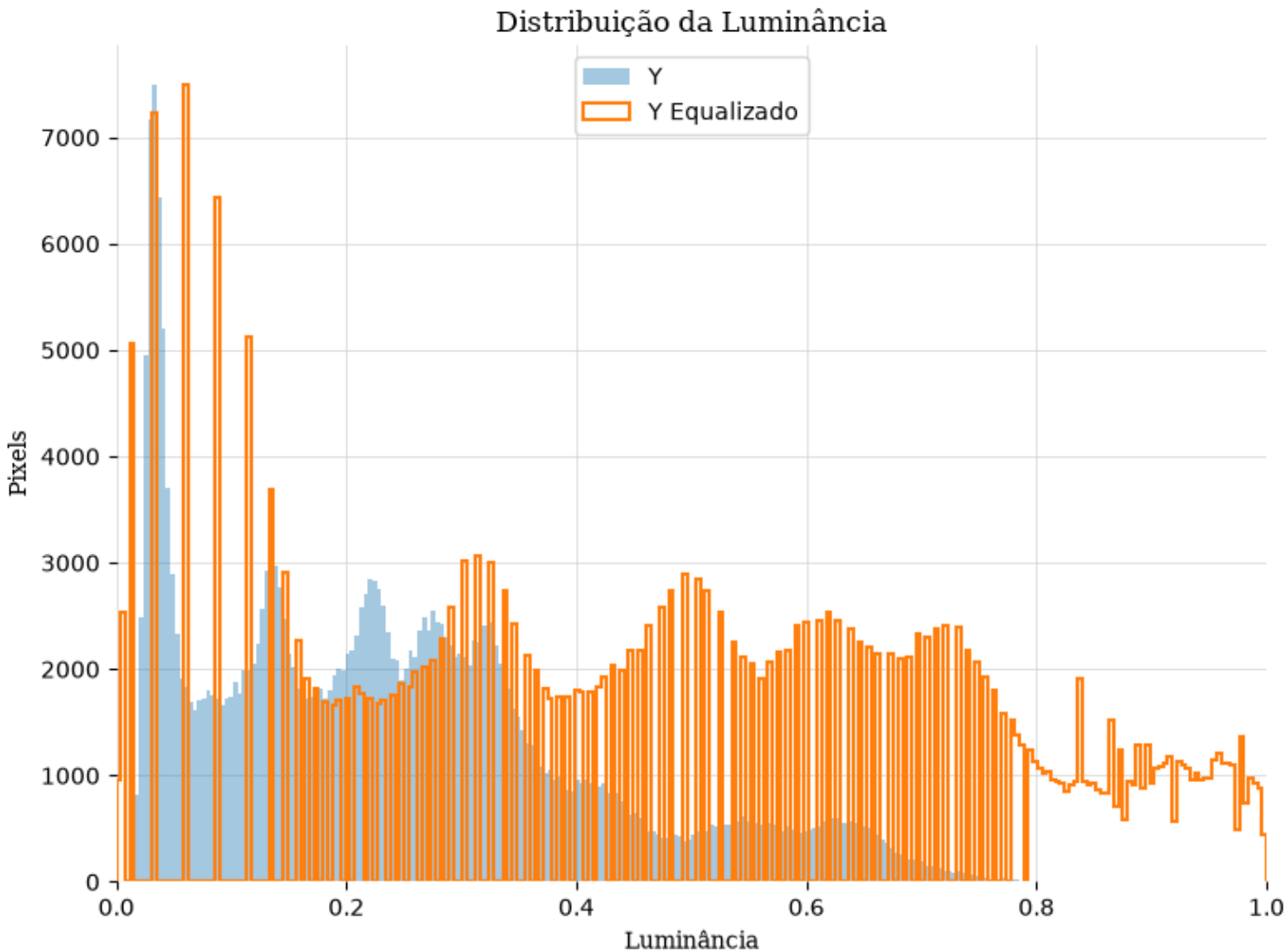


Espaços de cores

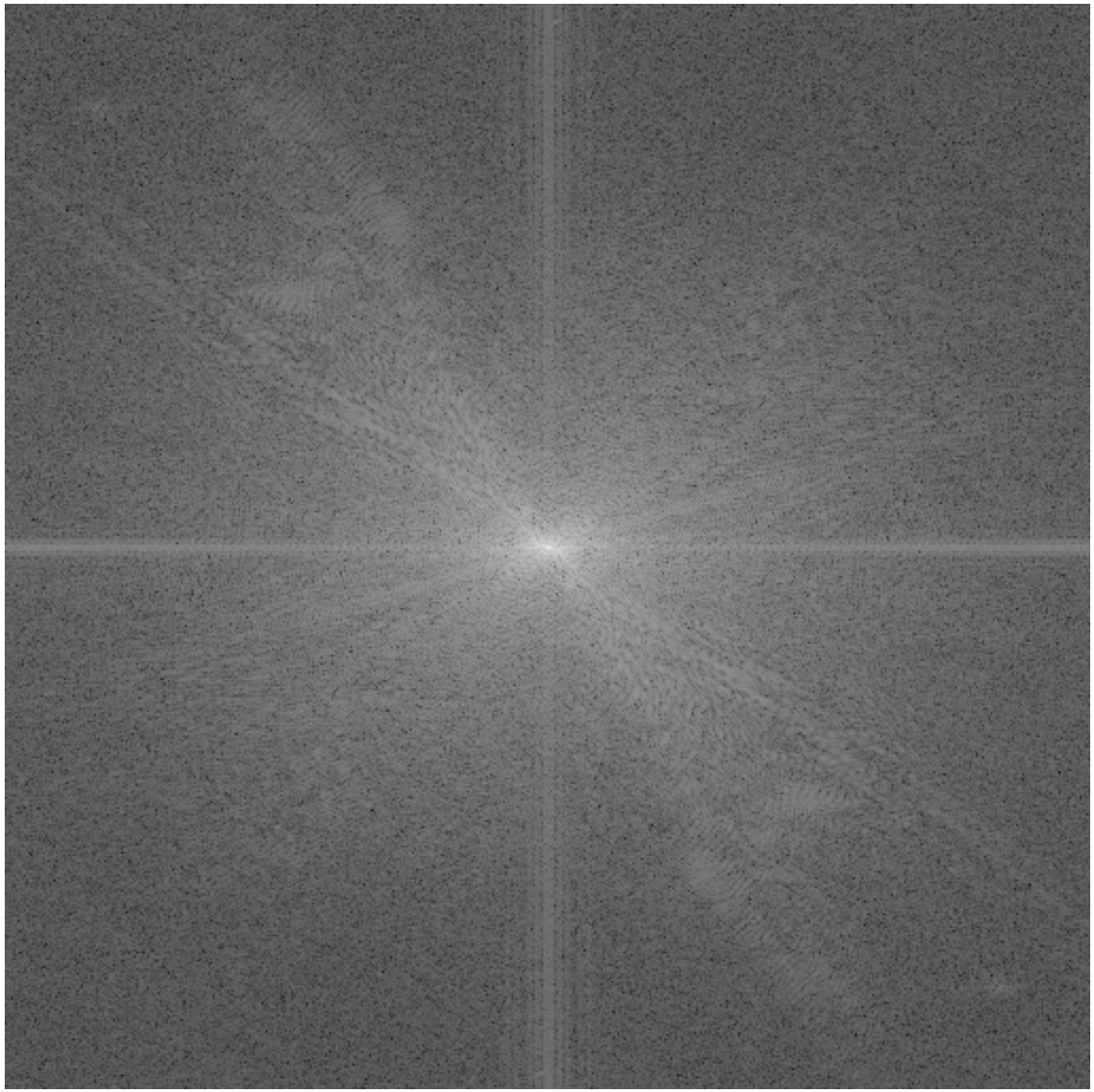




Balanceamento de contraste



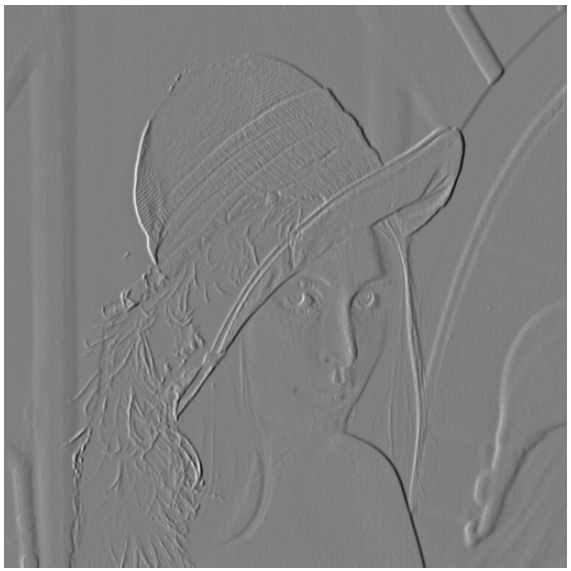
Análise espectral



Diferença das Gaussianas Baseada em Fluxo



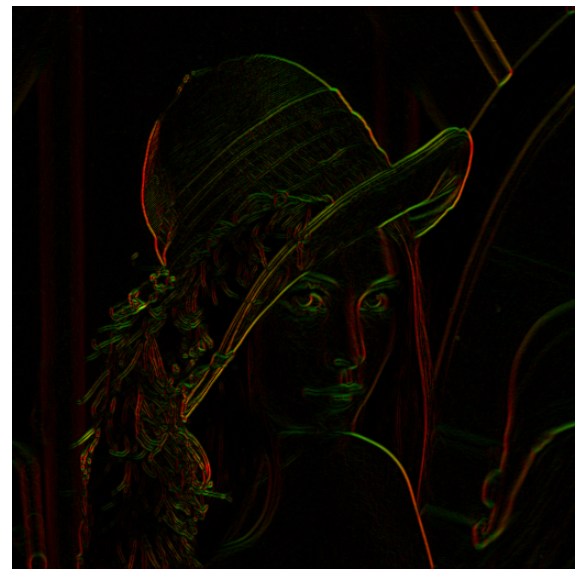
Diferença das Gaussianas Baseada em Fluxo



G_x



G_y



$(G_x, G_y, 0)$

Diferença das Gaussianas Baseada em Fluxo

$$S = \begin{bmatrix} G_x * G_x & G_x * G_y \\ G_x * G_x & G_y * G_y \end{bmatrix} = \begin{bmatrix} E & F \\ F & G \end{bmatrix}$$

Diferença das Gaussianas Baseada em Fluxo

